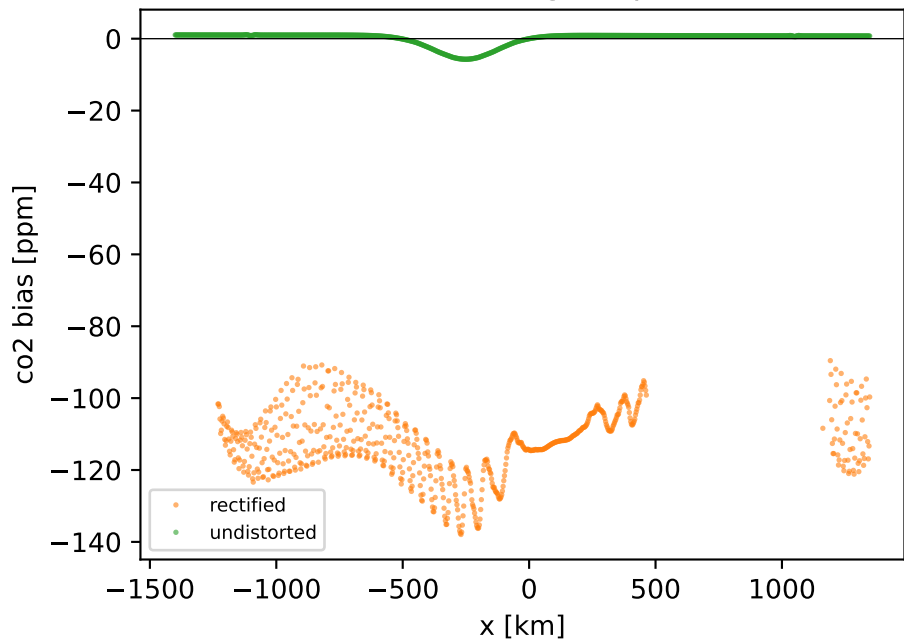
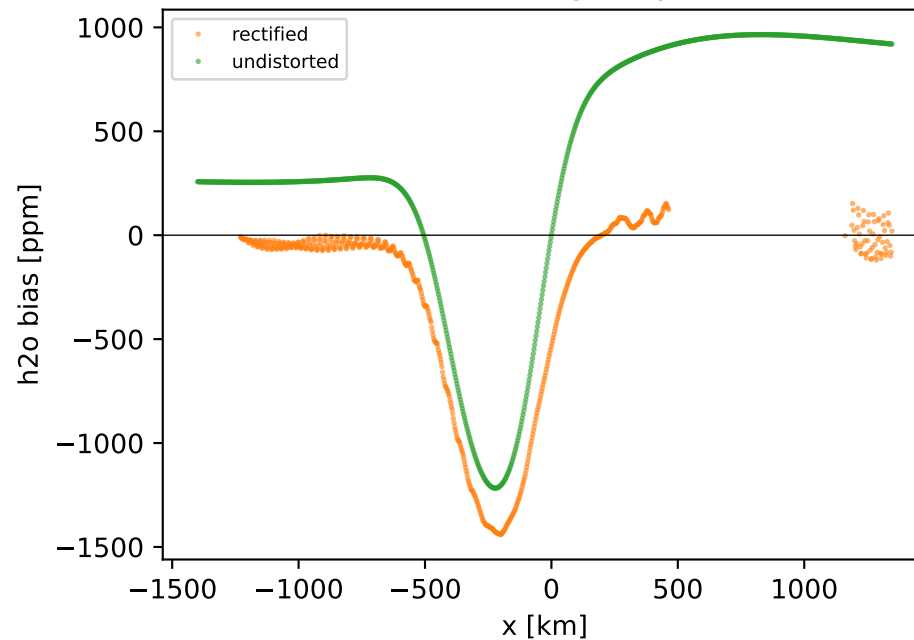


Joint FPA1 -- realistic, no noise (rectified/undistorted)

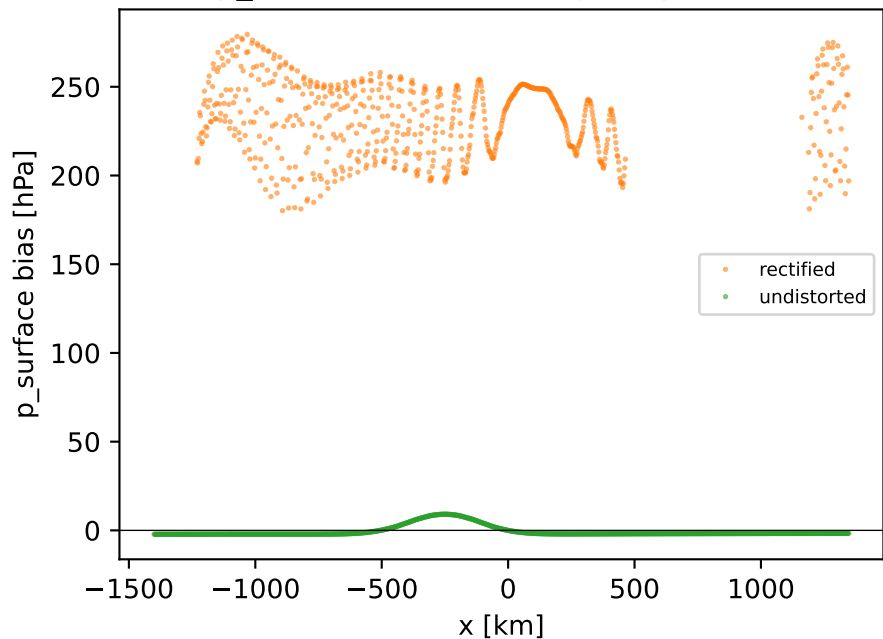
co2 bias vs along-slit position



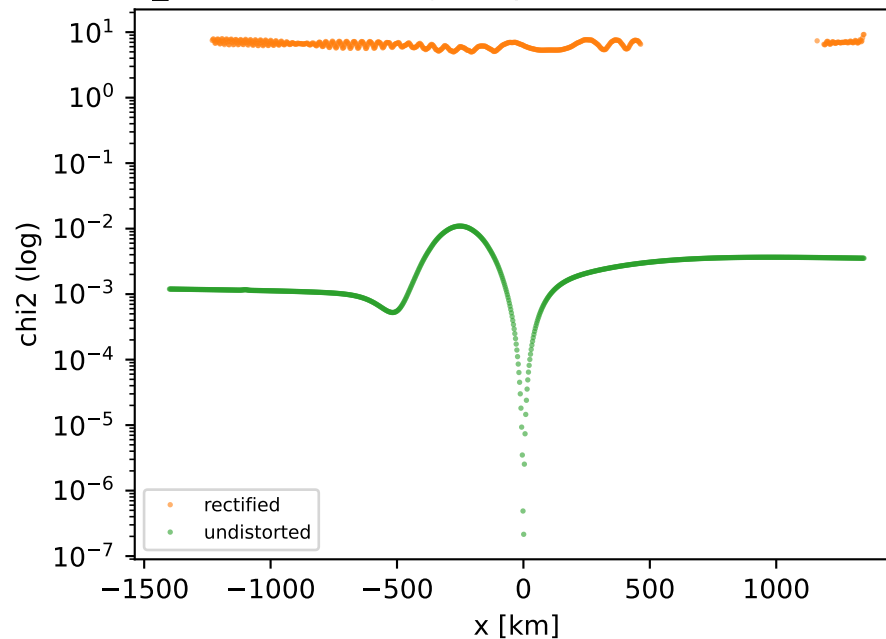
h2o bias vs along-slit position



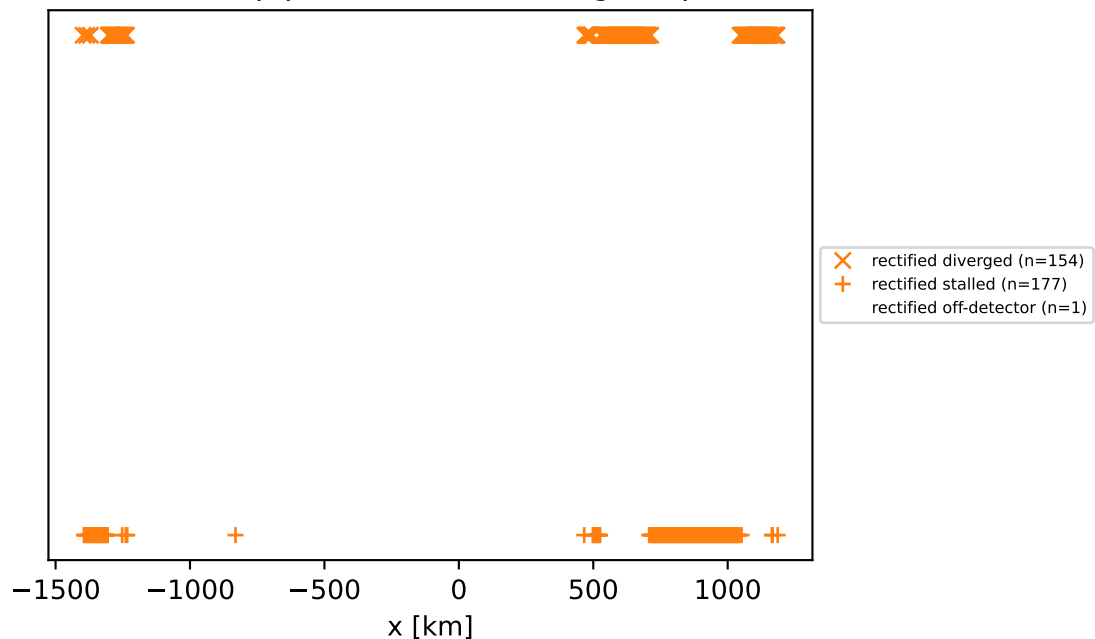
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

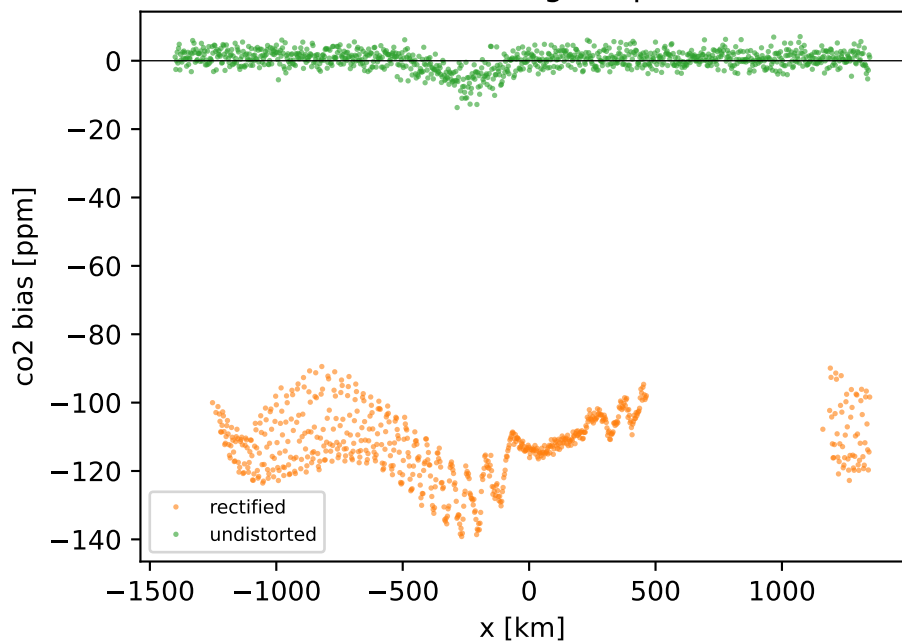


Where each pipeline fails, vs along-slit position

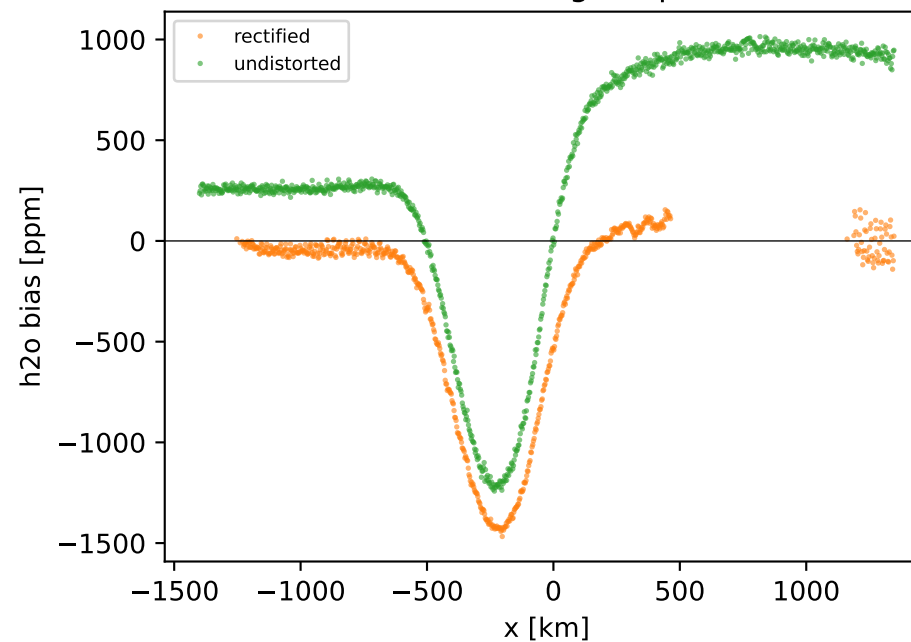


Joint FPA1 -- realistic, noise (rectified/undistorted)

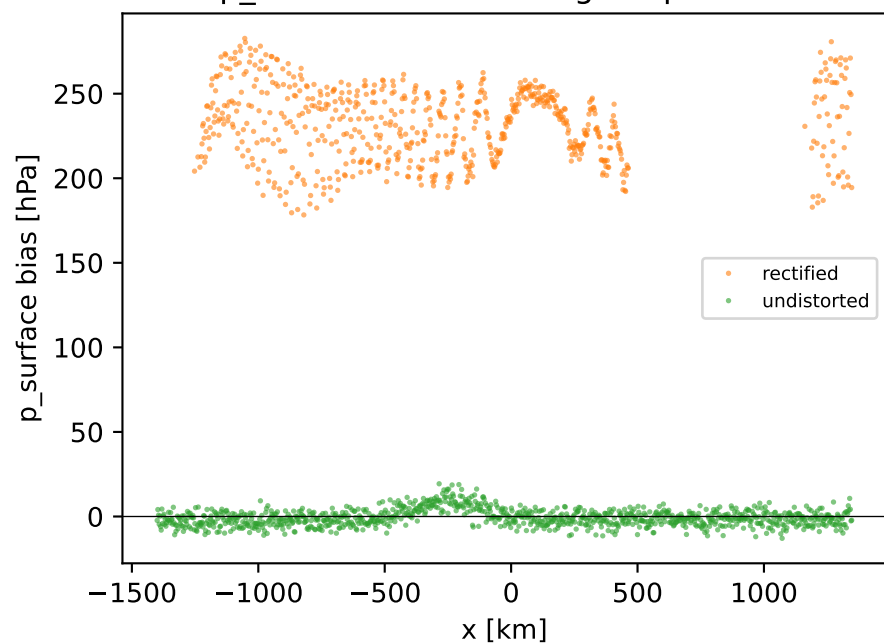
co2 bias vs along-slit position



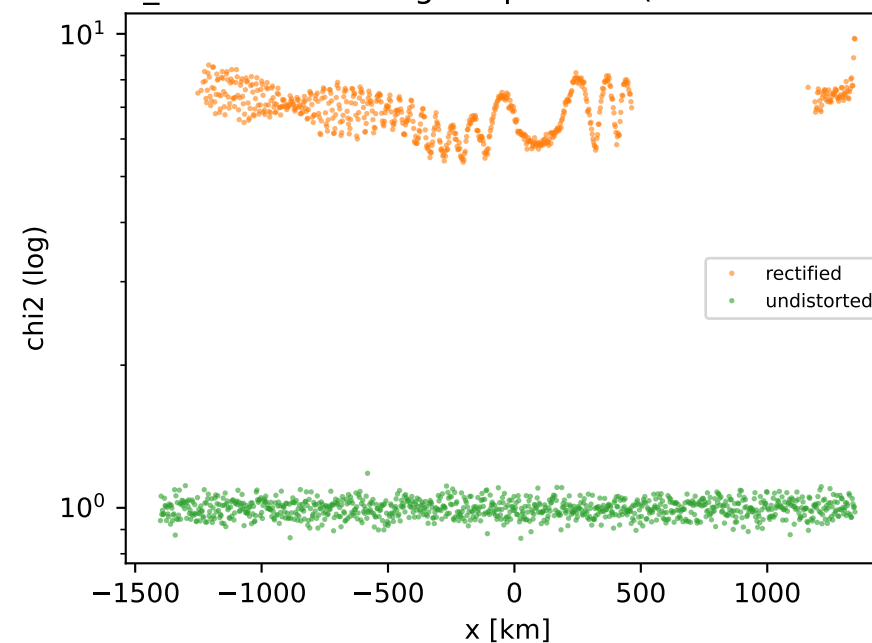
h2o bias vs along-slit position



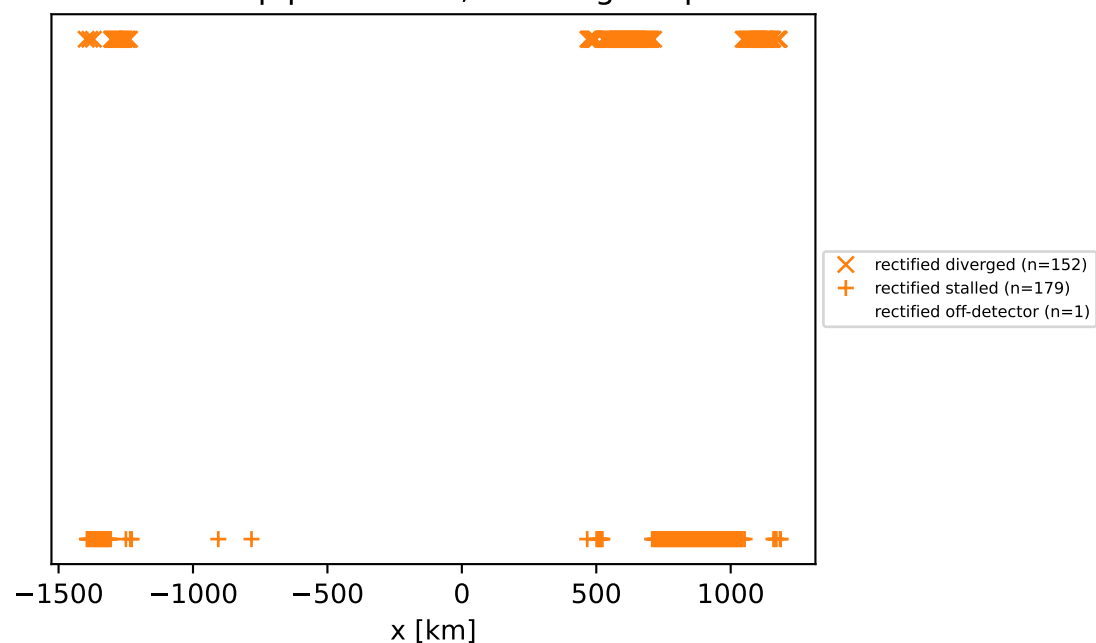
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

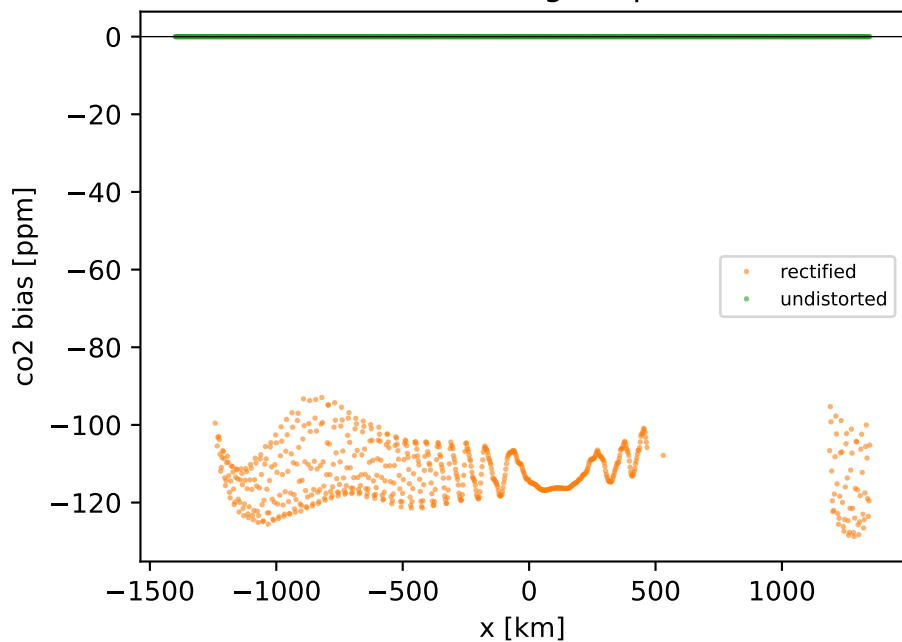


Where each pipeline fails, vs along-slit position

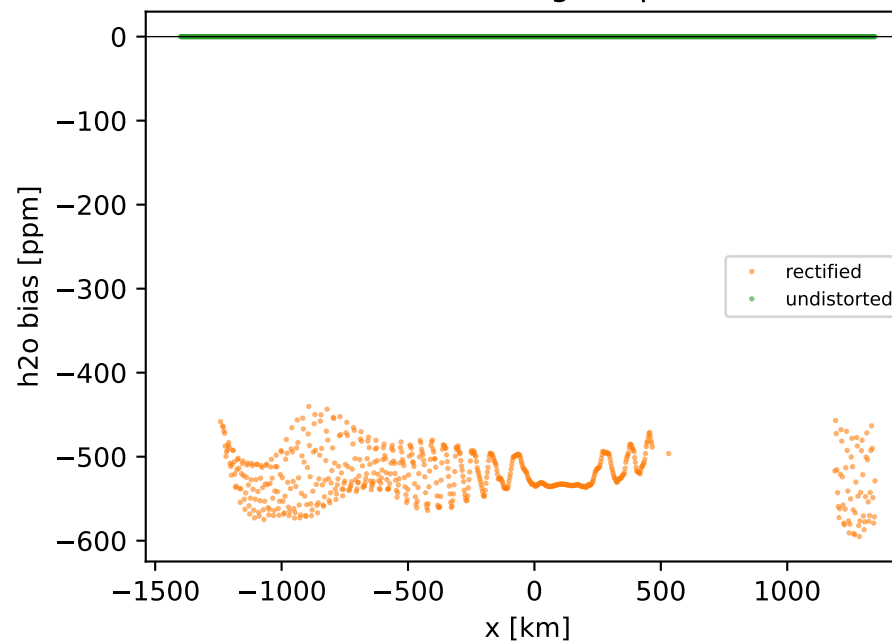


Joint FPA1 -- uniform, no noise (rectified/undistorted)

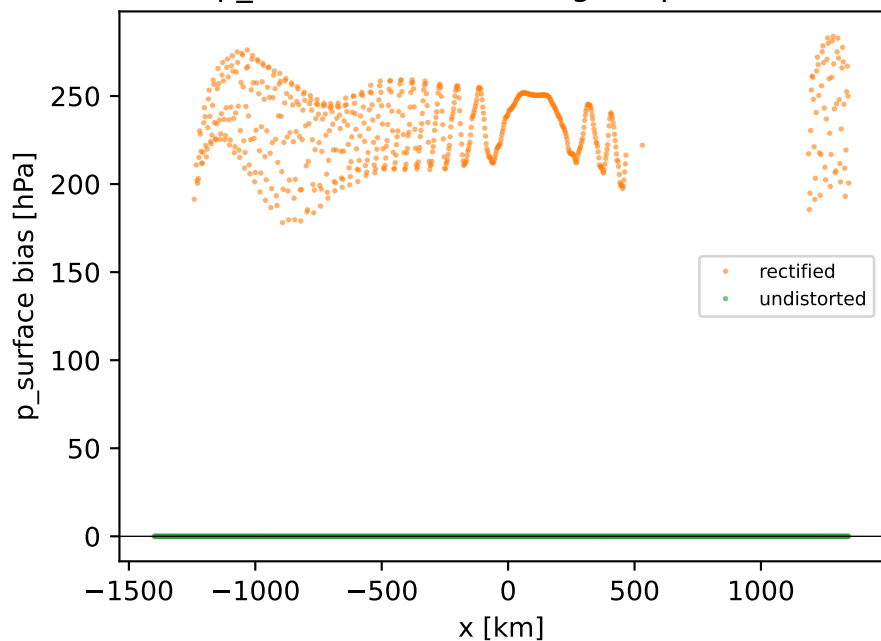
co2 bias vs along-slit position



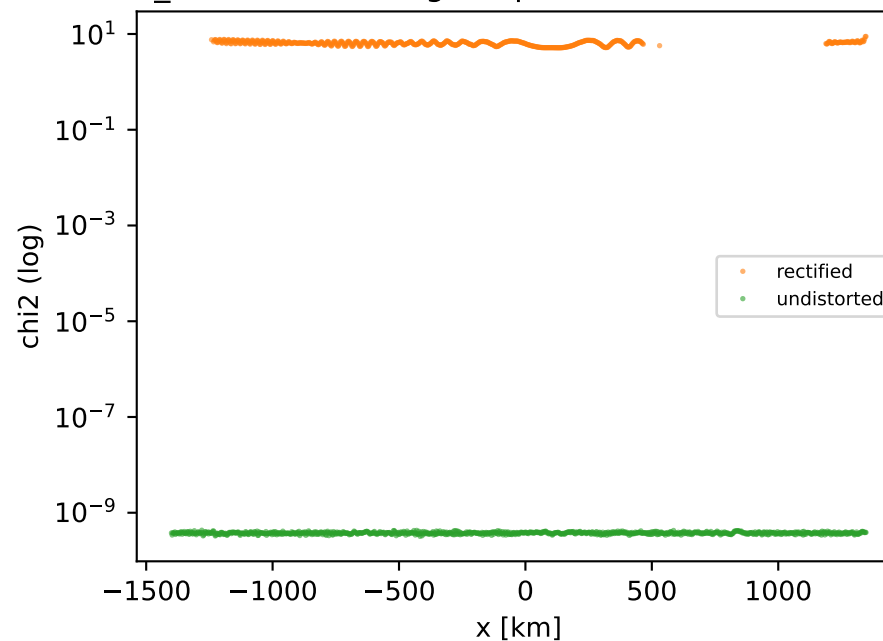
h2o bias vs along-slit position



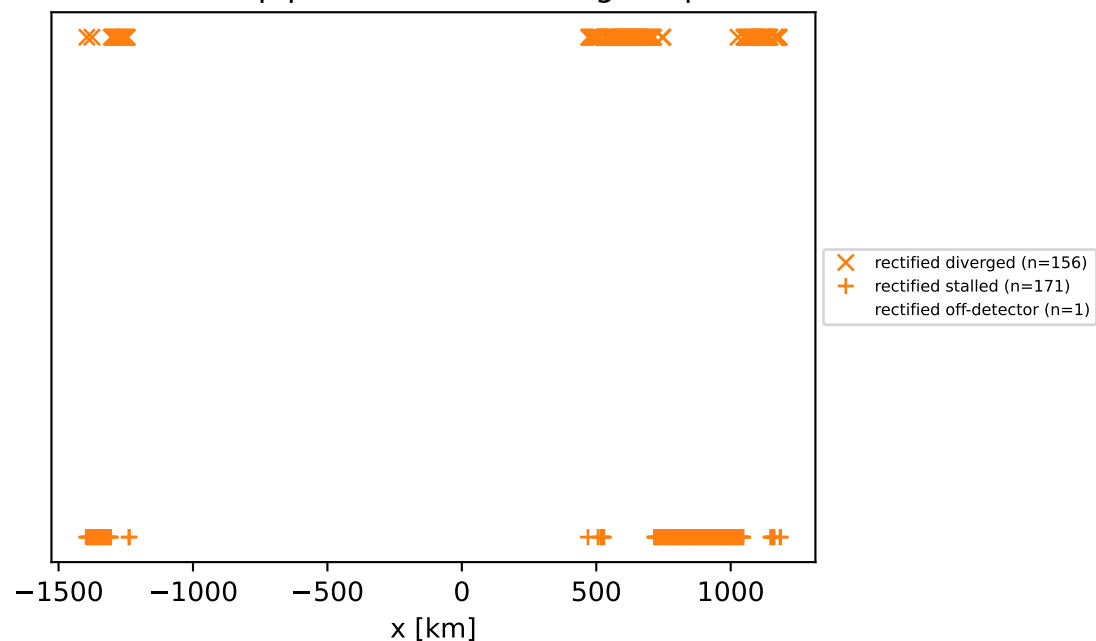
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

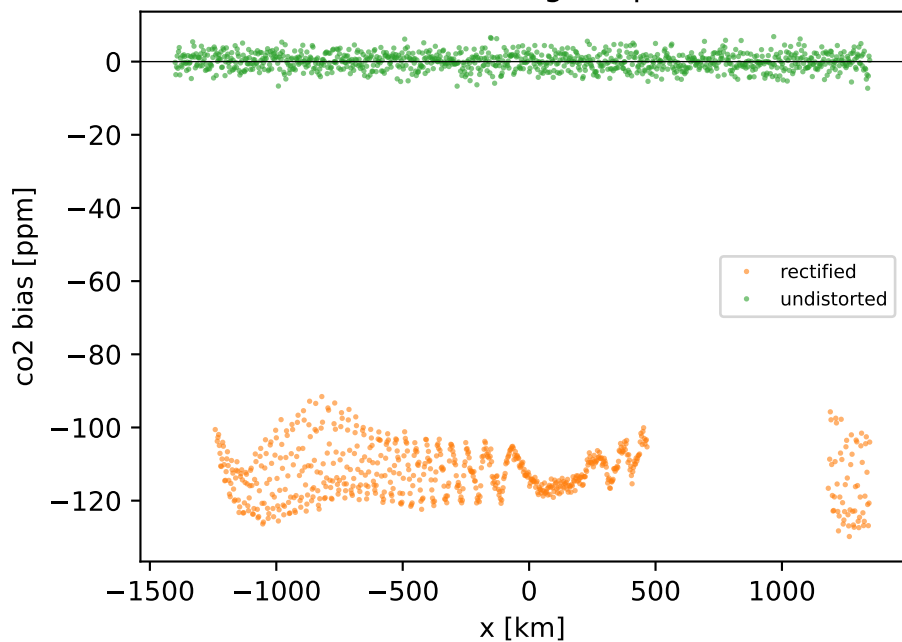


Where each pipeline fails, vs along-slit position

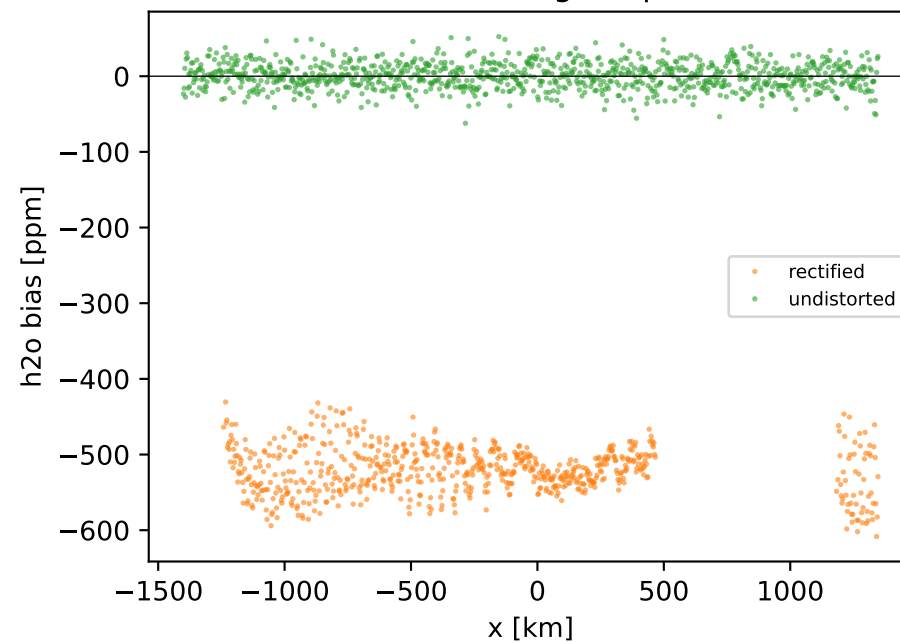


Joint FPA1 -- uniform, noise (rectified/undistorted)

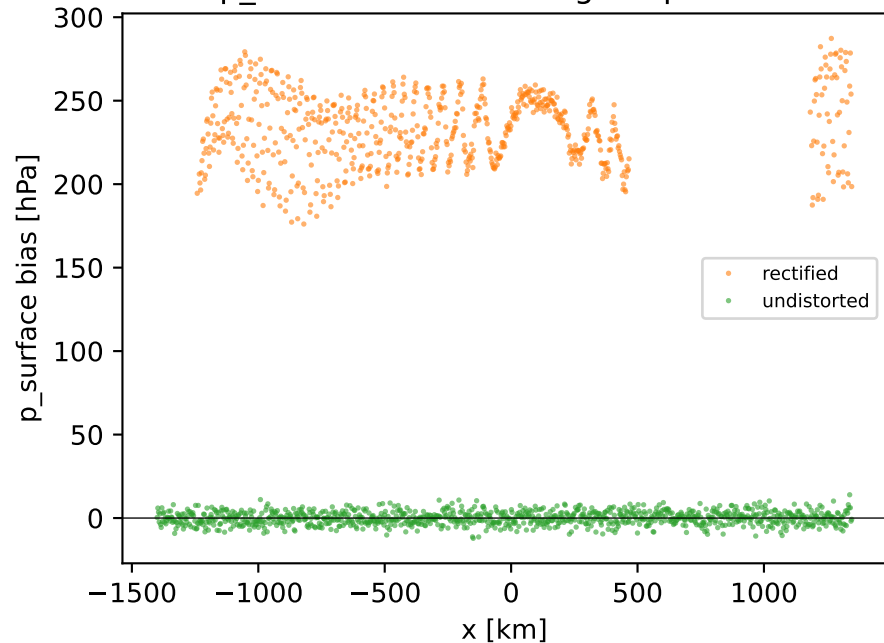
co2 bias vs along-slit position



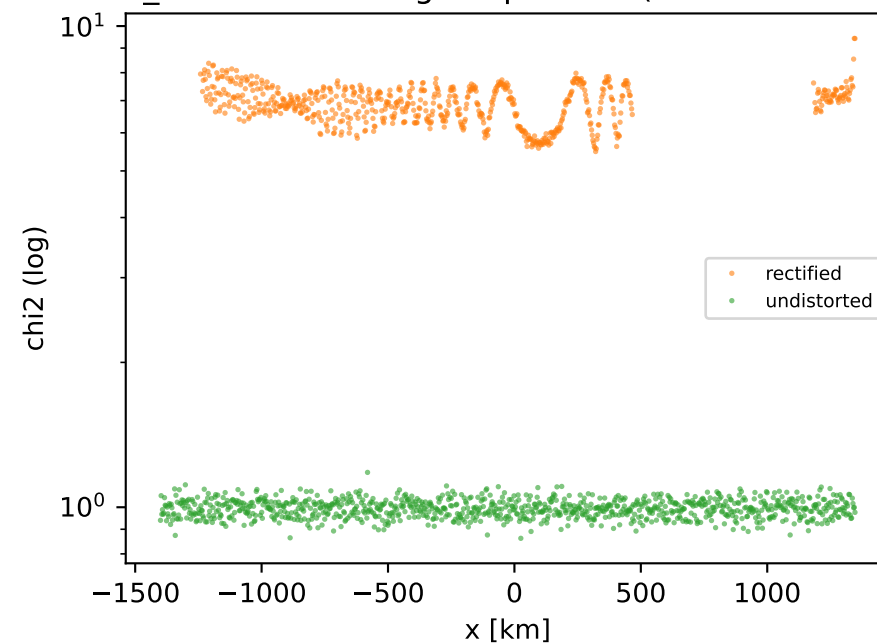
h2o bias vs along-slit position



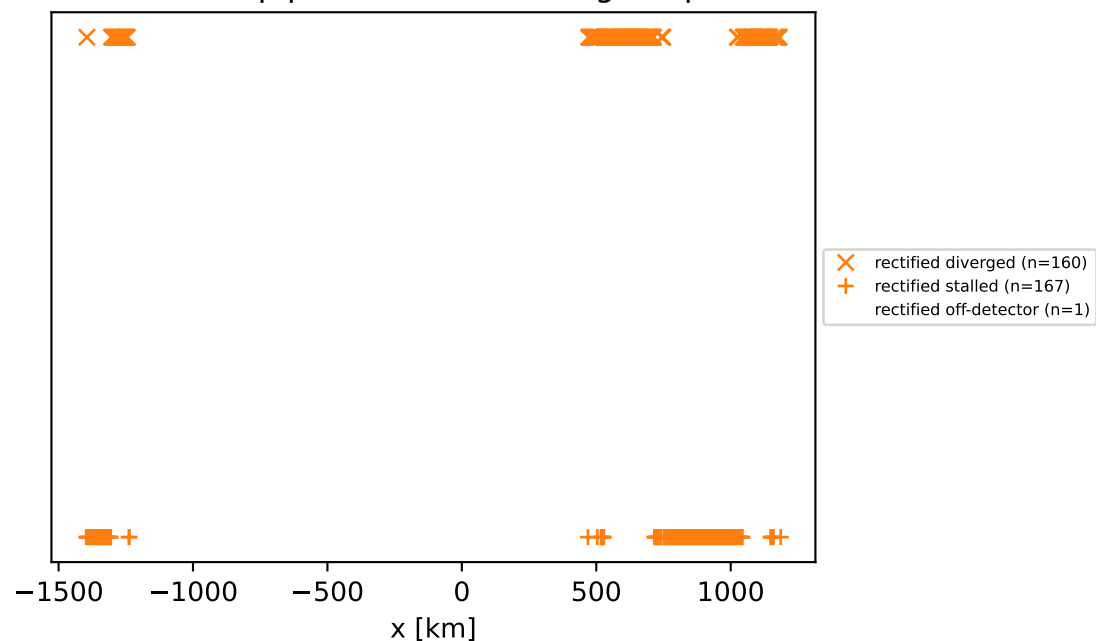
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

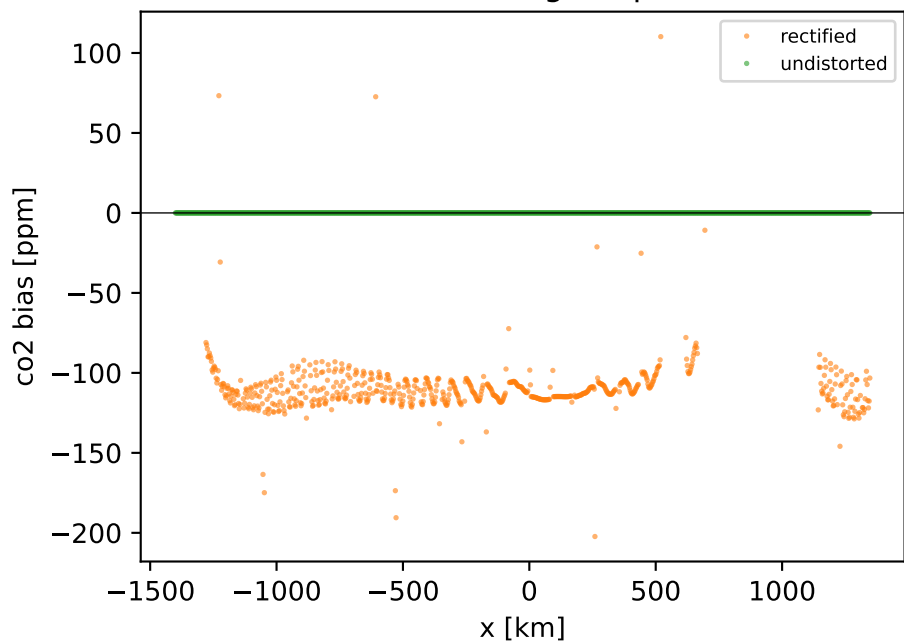


Where each pipeline fails, vs along-slit position

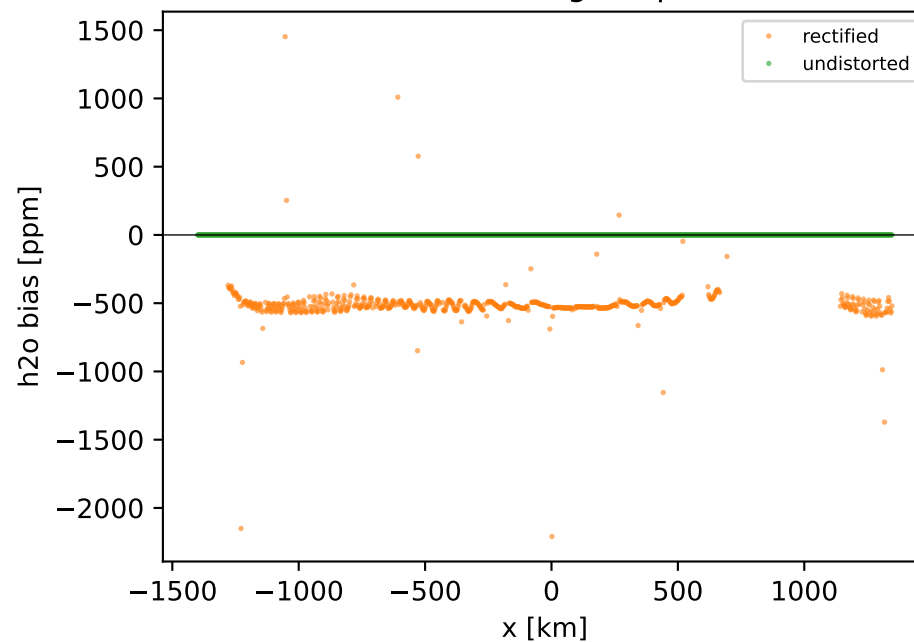


Joint FPA1 -- barcode, no noise (rectified/undistorted)

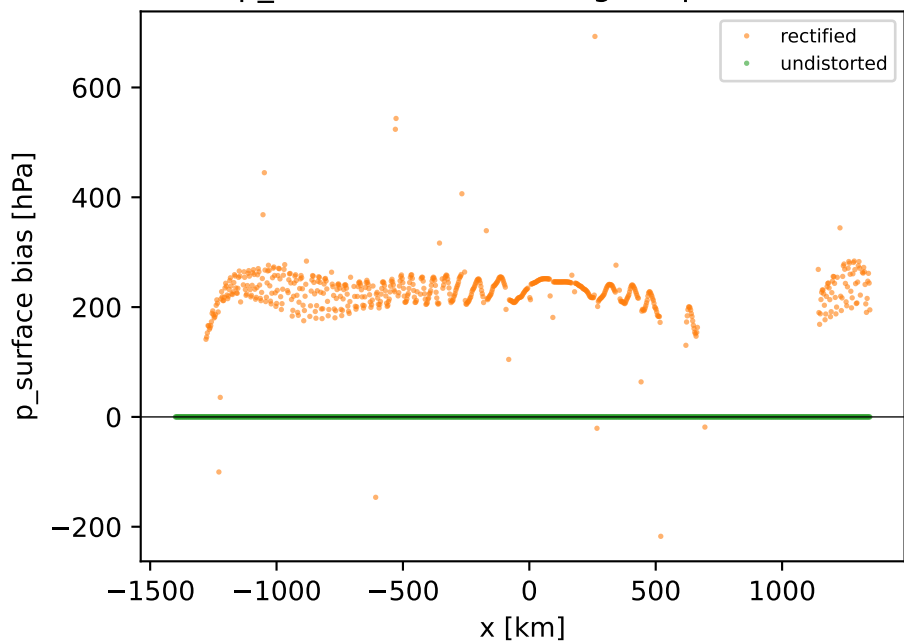
co2 bias vs along-slit position



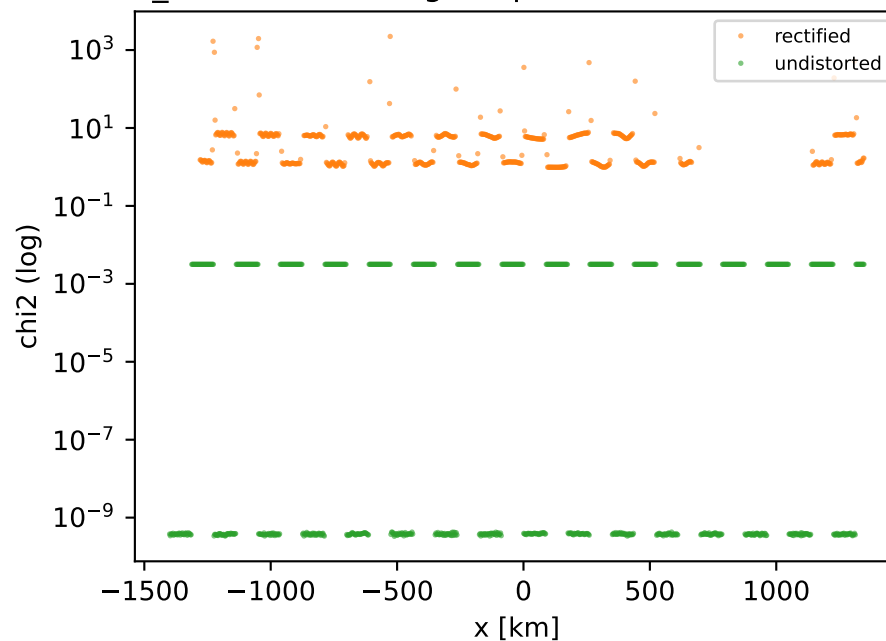
h2o bias vs along-slit position



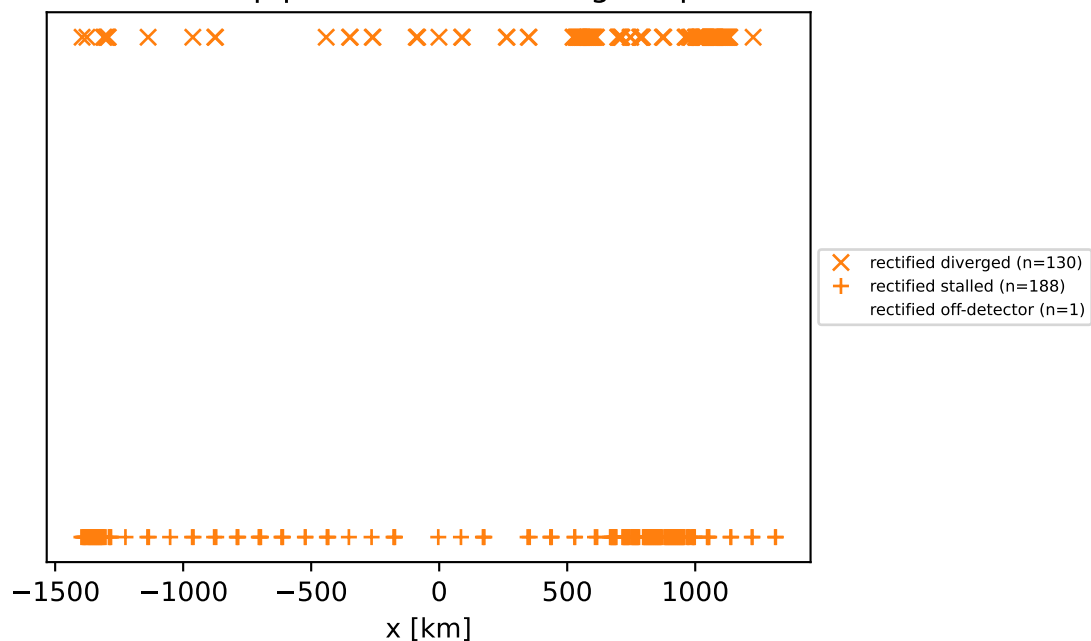
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

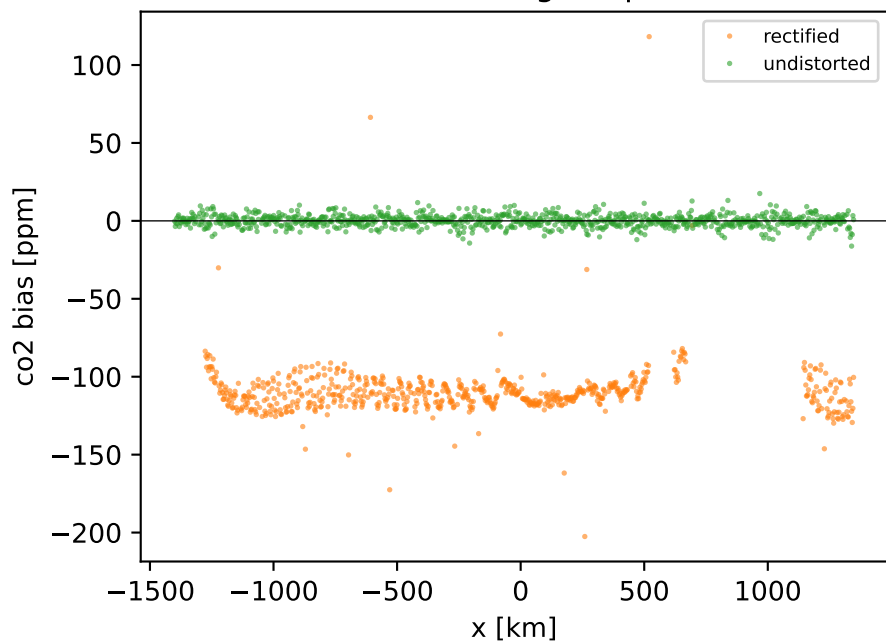


Where each pipeline fails, vs along-slit position

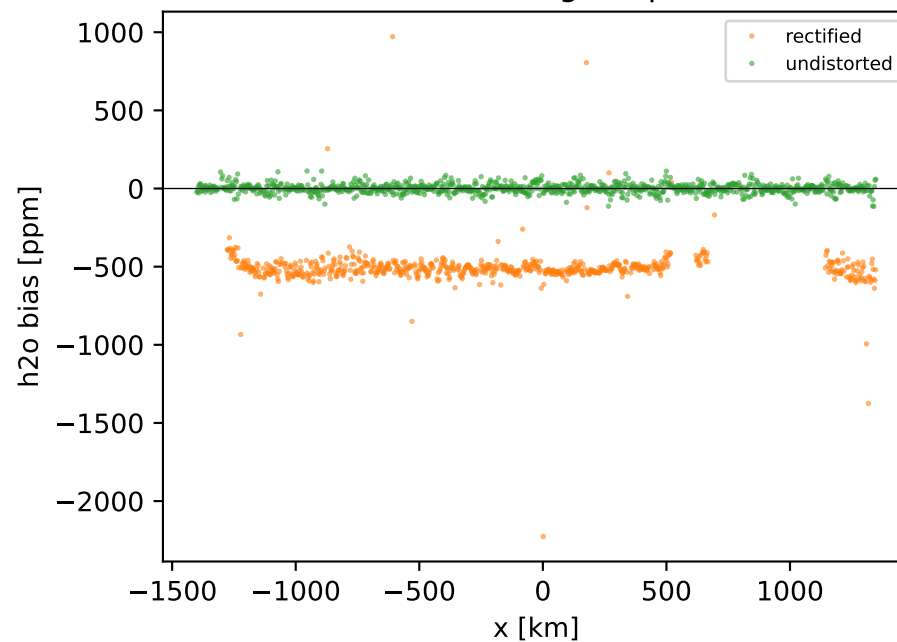


Joint FPA1 -- barcode, noise (rectified/undistorted)

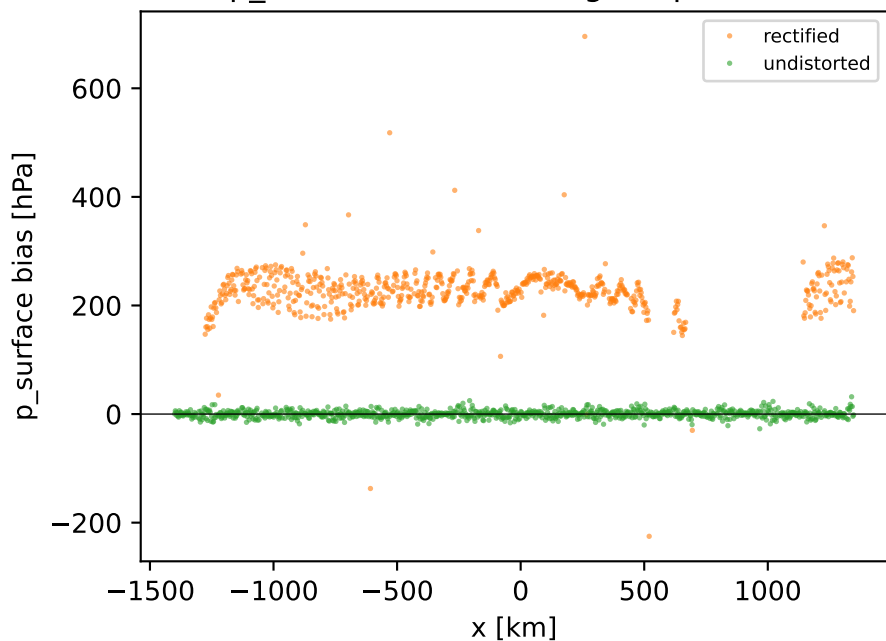
co2 bias vs along-slit position



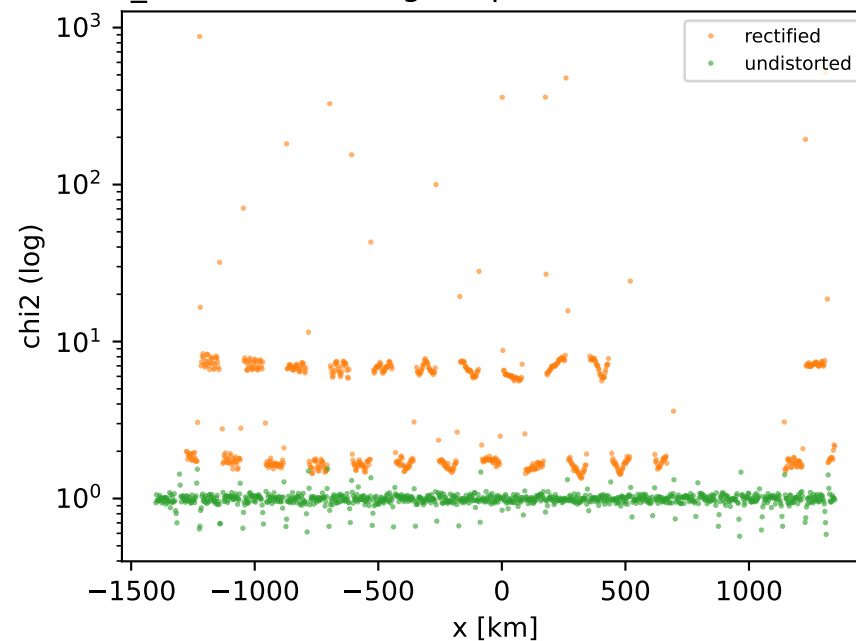
h2o bias vs along-slit position



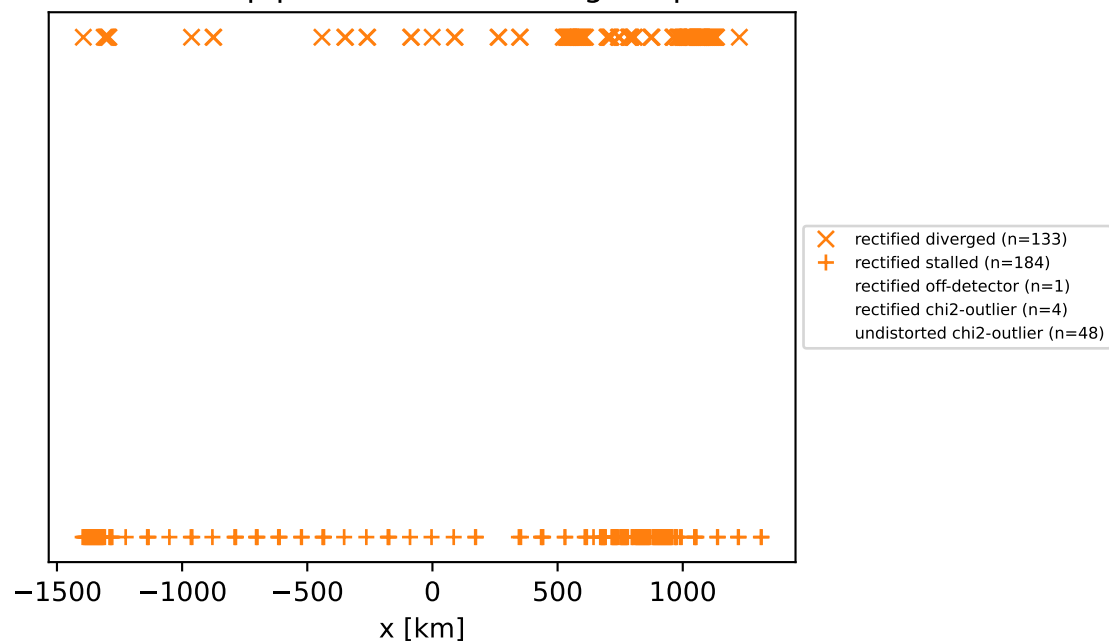
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)



Where each pipeline fails, vs along-slit position



FPA1 -- robust bias statistics across scenes/noise (rectified/undistorted)

scene	noise	quantity	pipeline	n	mean	std	median	MAD-sig	n_out
realistic	0	co2	rectified	692	-112.5	9.273	-112.3	8.522	0
realistic	0	h2o	rectified	692	-312	477	-60.82	142.7	78
realistic	0	p_surface	rectified	692	+232.1	21.08	+234	23.62	0
realistic	0	co2	undistorted	1024	+0.148	1.717	+0.8309	0.2005	154
realistic	0	h2o	undistorted	1024	+367.4	625.6	+274.7	887.2	0
realistic	0	p_surface	undistorted	1024	-0.5896	3.003	-1.834	0.379	159
realistic	1	co2	rectified	692	-112.5	9.376	-112.3	9.017	0
realistic	1	h2o	rectified	692	-311.8	476.6	-62.46	137.1	86
realistic	1	p_surface	rectified	692	+231.9	21.45	+233.9	23.93	0
realistic	1	co2	undistorted	1024	+0.1826	2.862	+0.482	2.507	0
realistic	1	h2o	undistorted	1024	+367.3	625.1	+282.1	870.3	0
realistic	1	p_surface	undistorted	1024	-0.5903	4.992	-0.9423	4.733	0
uniform	0	co2	rectified	696	-112.2	6.636	-112.9	6.709	0
uniform	0	h2o	rectified	696	-520.5	27.23	-524.4	25.63	0
uniform	0	p_surface	rectified	696	+233	20.54	+234.3	23.86	0
uniform	0	co2	undistorted	1024	-0.0005635	2.267e-05	-0.0005628	2.451e-05	0
uniform	0	h2o	undistorted	1024	-0.002689	0.0001374	-0.00269	0.0001341	0
uniform	0	p_surface	undistorted	1024	+0.0006725	4.124e-05	+0.0006703	4.555e-05	0
uniform	1	co2	rectified	696	-112.2	6.779	-112.6	6.946	0
uniform	1	h2o	rectified	696	-520.3	30.24	-521	29.66	0
uniform	1	p_surface	rectified	696	+232.9	21	+233.4	23.7	0
uniform	1	co2	undistorted	1024	+0.02591	2.301	-0.006723	2.315	0
uniform	1	h2o	undistorted	1024	+0.2565	17.38	+0.0308	17.35	0
uniform	1	p_surface	undistorted	1024	+0.009156	4.019	+0.04106	4.074	0
barcode	0	co2	rectified	705	-109.6	17.84	-111.5	7.596	11
barcode	0	h2o	rectified	705	-510.6	156	-517.7	29.73	16
barcode	0	p_surface	rectified	705	+227.2	47.77	+230.1	24.48	9
barcode	0	co2	undistorted	1024	-0.0007495	0.0001937	-0.0006074	0.0001291	0
barcode	0	h2o	undistorted	1024	-0.002627	0.0001748	-0.002633	0.0001661	0
barcode	0	p_surface	undistorted	1024	+0.001062	0.0004037	+0.0007541	0.0002345	0
barcode	1	co2	rectified	702	-109.9	15.99	-111.2	8.401	6
barcode	1	h2o	rectified	702	-508.5	125.6	-514.1	37.85	12
barcode	1	p_surface	rectified	702	+227.6	44.55	+228.7	26.54	6
barcode	1	co2	undistorted	976	+0.04373	3.832	-0.02095	3.368	0
barcode	1	h2o	undistorted	976	+0.9155	30.05	+0.261	23.72	0
barcode	1	p_surface	undistorted	976	+0.03428	6.67	+0.06534	5.928	0